

OMAIMA IRFAN

Software Engineer — AI & Backend Development

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SUMMARY

Software Engineer specializing in AI-driven and full-stack web applications, with hands-on experience building backend systems using Python, FastAPI, and Angular. Skilled in RAG architectures, LLM integration, vector databases, and ETL pipelines. Seeking an internship or entry-level role in Backend, AI/ML, or Full-Stack Development.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, SQL
- **Backend & APIs:** FastAPI, REST API Design, RAG Architecture, Document Processing (PDF/PPT/DOCX)
- **Frontend:** Angular, HTML, CSS, JavaScript
- **AI / Machine Learning:** OpenCV, Emotion & Image Recognition, LLM Integration, Embeddings, Semantic Search
- **Databases & Storage:** MySQL, ChromaDB, pgvector
- **Other:** ETL Pipelines, Git, Arduino/Embedded Basics, Interaction Design, Visual Design

PROFESSIONAL EXPERIENCE

Software Engineer (AI/Backend) — TeachAssist

Aug 2025 – Present

Gap Analyzer & AI-Based Exam Generation Platform — [GitHub repo](#)

- Built an OBE-aligned gap analysis system that automatically detects weak learning outcomes from exam results, helping instructors target remediation efficiently.
- Designed and developed backend services in FastAPI to automate processing of assessment papers and Excel-based academic data, cutting manual review time.
- Engineered an AI-based exam and assignment generation system using Angular and FastAPI (Python), integrating a Retrieval-Augmented Generation (RAG) pipeline with LLMs for automated quiz and paper creation.
- Implemented end-to-end document processing (PDF, PPT, DOCX) including text extraction and embedding generation for semantic search.
- Used ChromaDB and pgvector for vector storage, and built optimized REST APIs to improve retrieval performance and response times.
- Generated AI-assisted academic improvement recommendations for students based on gap analysis output.

PROJECTS

Emotion Detection System — Machine Learning

Oct 2024 – Jan 2024

Python, OpenCV

- Developed a facial emotion recognition system using Python and OpenCV to classify emotional states from image and video input in real time.
- Applied image preprocessing techniques (face detection, normalization, feature extraction) to improve model input quality and classification accuracy.
- Evaluated model performance across multiple emotion categories and documented results to support further tuning and dataset expansion.

Restaurant Ordering Web Application

Oct 2024 – Jan 2024

HTML, CSS, JavaScript, MySQL

- Developed a full-stack restaurant ordering system with cart, checkout, and order-tracking functionality.
- Designed and implemented a relational MySQL database schema for menu items, orders, and order management.
- Built a responsive, interactive UI with custom animations to improve the customer ordering experience.

Traffic Light Control System — Computer Organization & Assembly Language

Arduino

- Built an Arduino-based traffic light control system, programming timed signal sequencing and simulating real-world intersection logic.

Student Record Management System — Database Systems

MySQL

- Designed and implemented a relational database for managing student records, including schema design and query operations for data retrieval and updates.

EDUCATION

Bachelor of Science in Software Engineering

2022 – 2026

UIT University

- Focused coursework in Artificial Intelligence, Web Development, and Data Engineering.
- Core subjects: Data Structures & Algorithms, Database Systems, Machine Learning.

Intermediate (Pre-Engineering)

2020 – 2022

SRE Majeed College